Setup Instructions for MySQL

Setup Guide: CMS Data Governance Project (MySQL on Windows)

This guide provides step-by-step instructions for downloading and running the CMS Data Governance project on a Windows machine using a local MySQL database.

1. Download the Project (Clone)

First, download the code from GitHub to your local machine using the command line.

1. Open the **Command Prompt** or **PowerShell**.
2. Navigate to the folder where you want to store the project (e.g., your Documents folder):

   DOS

   ```
   cd Documents
   ```

3. Run the following command to clone the repository:

   DOS

   ```
   git clone https://github.com/qsanders/CMS_Data_Governance_Project.git
   ```

4. Move into the newly created project folder:

   DOS

   cd CMS_Data_Governance_Project

2. Set Up a Virtual Environment

A virtual environment ensures the required Python packages do not interfere with other projects on your system.

1. Create a virtual environment named `venv`:

   DOS

   ```
   python -m venv venv
   ```

2. Activate the virtual environment:

```
DOS
venv\Scripts\activate
```

(You will know this worked when you see `(venv)` at the beginning of your command line prompt).

3. Install Dependencies & the MySQL Driver

Install the base project requirements and the specific driver needed to communicate with MySQL.

1. Install the base packages:

```
DOS
pip install -r requirements.txt
```

2. Install the MySQL connection driver (`pymysql`):

```
DOS
pip install pymysql
```

4. Configure Your Environment Variables

The script uses a `.env` file to securely load your local database credentials.

1. In the root directory of the `CMS_Data_Governance_Project` folder, create a new plain text file and name it exactly: `.env`
2. Open this file in a text editor (like Notepad or VS Code) and add the following parameters. Adjust the username and password to match your local MySQL setup:

```
Plaintext

DB_HOST=localhost
DB_PORT=3306
DB_NAME=cms_data_governance
DB_USER=your_mysql_username
DB_PASS=your_mysql_password
```

(Note: Ensure that the database specified in `DB_NAME` already exists in your local MySQL instance before moving forward).

5. Update the Database Connection String

Because this project uses SQLAlchemy, you just need to make a minor one-line change to the script to switch it from PostgreSQL to MySQL.

1. Open the following file in your code editor: `code_scripts/raw_cms_DE_SynPUF_data_loader.py`
2. Scroll to the `get_database_engine()` function (around line 50).
3. Replace the entire function with the following block to use the MySQL syntax:

```python
def get_database_engine():
    """Builds the database connection and handles connection errors."""
    db_user = os.getenv("DB_USER")
    db_pass = os.getenv("DB_PASS")
    db_host = os.getenv("DB_HOST")
    db_port = os.getenv("DB_PORT", "3306")  # Set default port fallback to MySQL (3306)
    db_name = os.getenv("DB_NAME")


 # Updated connection string specifically for MySQL using pymysql
   url = f"mysql+pymysql://{db_user}:{db_pass}@{db_host}:{db_port}/{db_name}"


   return create_engine(url)
```

4. Save the file.

6. Run the Script

With your virtual environment active, your `.env` file saved, and the script updated, you are ready to load the data.

Run the following command in your terminal:

DOS

```
python code_scripts/raw_cms_DE_SynPUF_data_pull.py
```

The script will now pull the raw filesdown from the CMS website. You can check the `raw_cms_DE_SynPUF_data_pull.py.log` file in the same folder for real-time progress and output metrics.